

SOLUTION

FINAL EXAMINATION REVISION EXERCISES

BSTA001 - BUSINESS STATISTICS

IMPORTANT INFORMATION REGARDING YOUR FINAL EXAM

Time allowed: 3 hours plus 10 minutes reading time

Number of Questions:  **Question 1:** 20 multiple-choice questions

 **Question 2-13:** Short-answer questions

Instructions: **You must attempt ALL questions**

For any numerical calculations, show your workings **correct to 4 decimal places** unless stated otherwise.

Materials Allowed: Electronic calculators allowed.

ONE (1) two-sided A4 sheets with handwritten or typed notes allowed.

Question 1: Multiple choice Questions

☛ Draw a circle around the letter corresponding to the correct answer. Circle **ONE** answer only.

1. A student achieves an 82 on the first test in a statistics course. From this, she assumes that her average at the end of the semester (after other tests) will be about 82. This is an example of
 - A. descriptive statistics
 - B. inferential statistics**
 - C. nonparametric statistics
 - D. wishful thinking
2. The Jacaranda Swimming Pool Company wants to determine the average number of years it takes before a major repair is required on one of their pools. The manager of the company asks Rick Johnson, company accountant, to randomly contact fifty families who purchased Jacaranda pools in the past ten years and determine how long it was in each case before a major repair was needed. The information will be used to estimate the average number of years before a major repair is required for all pools sold by Jacaranda. The average, based on the data gathered from the fifty families, can best be described as a
 - A. statistic**
 - B. sample
 - C. population
 - D. parameter
3. Which of the following is NOT a commonly used measure of variability?
 - A. variance
 - B. standard deviation
 - C. mode**
 - D. interquartile range
4. A graphical representation of a frequency distribution is called a
 - A. stem and leaf plot
 - B. ogive
 - C. pie chart
 - D. histogram**
5. The list of all elementary events for an experiment is called
 - A. the sample space**
 - B. the exhaustive list
 - C. the population space
 - D. the event union

6. Jessica Salas, Managing Director of Salas Products, is reviewing the warranty policy for her company's new model of car batteries. Life tests performed on a sample of 100 batteries indicated: (1) an average life of 75 months, (2) a standard deviation of 5 months, and (3) battery life is approximately normally distributed. What percentage of the batteries will fail within the first 65 months of use?
- A. Approximately 0.5%
 - B. Approximately 2.5%**
 - C. Approximately 1%
 - D. Approximately 5%
7. Mark Sims is exploring the characteristics of stock market investors. He found that 60% of all investors have a net worth exceeding \$1,000,000; 20% of all investors use an online brokerage; and 10% of all investors have a net worth exceeding \$1,000,000 and use an online brokerage. An investor is selected randomly, and E is the event net worth exceeds \$1,000,000, and O is the event uses an online brokerage. $P(E \cup O) = \dots\dots\dots$
- A. 0.17
 - B. 0.50
 - C. 0.70**
 - D. 0.80
8. If x is a binomial random with $n = 8$ and $p = 0.6$, what is the probability that x is equal to 5?
- A. 0.625
 - B. 0.209
 - C. 0.300
 - D. 0.279**
9. If lambda λ is 4 occurrences per five minute time interval, the probability of getting 3 occurrences over a five minute interval is $\dots\dots\dots$
- A. 0.1954**
 - B. 0.0183
 - C. 0.2237
 - D. 0.1680
10. If x is uniformly distributed over the interval 20 to 30, inclusively ($20 \leq x \leq 30$), then the standard deviation of this distribution is $\dots\dots\dots$
- A. incalculable
 - B. 2.89**
 - C. 8.33
 - D. 0.833

11. Financial analyst Ben Taimana needs a sample of 100 securities listed on the New Zealand Stock Exchange. In the current issue of the NZ Investing Journal, 2531 securities are listed in the Composite Transactions, an alphabetical listing of all securities traded on the previous business day. Ben uses a table of random numbers to select 100 numbers between 1 and 2531. His sample is a
- A. *simple random sample*
- B. stratified sample
- C. systematic sample
- D. quota sample
12. Let z be a normal random variable with mean 0 and standard deviation 1. Use the normal tables to find $P(1.3 < x < 2.3)$.
- A. 0.4032
- B. 0.9032
- C. *0.0861*
- D. 0.4893
13. The EPA has reported that the average fuel cost for a particular type of car is \$800 with a standard deviation of \$80. Fuel cost is assumed to be normally distributed. If one of these cars is randomly selected, what is the probability that the fuel cost for this car exceeds \$900?
- A. 0.3944
- B. *0.1056*
- C. 0.8944
- D. 0.6056
14. Multiple regression analysis produced the following tables.

	SS	df	MS	F	Significance F
Regression	3340.7	4	849.4	3.65	0.029
Residual	3493.1	15	232.9		
Total	6890.8	19			

What conclusion can be reached about this regression model from this output?

- A. At $\alpha = 0.01$, at least one of the regression coefficients is different from zero.
- B. At $\alpha = 0.001$, at least one of the regression coefficients is different from zero.
- C. *At $\alpha = 0.05$, at least one of the regression coefficients is different from zero.*
- D. None of the regression coefficients are different from zero.
15. When a range of values is used to estimate a population parameter, it is called
- A. a point estimate
- B. a statistical parameter
- C. a range estimate
- D. *an interval estimate*

16. James Weepu, Human Resources Manager with Auckland First Bank (AFB), is reviewing the employee training programs of AFB branches. He plans to use a 90% confidence interval estimate of mean training time of tellers and is willing to accept an error of 1 hour; previous studies indicated a standard deviation of 5 hours. The sample size should be at least

A. 4
B. 16
C. 34
D. 68

17. Consider the following null and alternative hypotheses.

$$H_o : \mu \geq 67$$

$$H_a : \mu < 67$$

These hypotheses

A. indicate a one-tailed test with a rejection area in the right tail
B. indicate a one-tailed test with a rejection area in the left tail
C. indicate a two-tailed test
D. are established incorrectly

18. Powell's Pharmacy Ltd operates a regional chain of 120 pharmacies. Each pharmacy's floor plan includes a greeting card section which is relatively isolated. Sandra Ronaki, Marketing Manager, feels that the level of lighting in the greeting card section may affect sales in this area. She chooses three levels of lighting (soft, medium and bright) and randomly assigns six pharmacies to each lighting level. Sandra's null hypothesis is

A. $\mu_1 = \mu_2 = \mu_3$
B. $\mu_1 \neq \mu_2 \neq \mu_3$
C. $\mu_1 \geq \mu_2 \geq \mu_3$
D. $\mu_1 \leq \mu_2 \leq \mu_3$

19. A manager wishes to predict the annual cost (y) of a car based on the number of kilometres (x) driven. The following model was developed: $y = 1,550 + 0.36x$. If a car is driven 20,000 km, the predicted cost is

A. 7,200
B. 8,750
C. 5,650
D. 2,270

Short-answer Questions

- ☛ Write the answer to the question in the space provided.
- ☛ For any numerical calculations, show your workings correct to 4 decimal places unless stated otherwise.

Question 2:

The following, supplied by Marketing Intelligence Service, is a list of companies with the most new products in the year 2008.

Company	Number of new products
Avon Products	768
L'Oreal	429
Unilever U.S.	323
Revlon	306
Garden Botanika	286

- (a) Find the mean, mode and median.

☛ Mean: $\bar{x} = 422.4$

☛ Mode: Nil

☛ Median: **323**

- (b) Find the sample variance and standard deviation.

Company	x	$x - \bar{x}$	$(x - \bar{x})^2$
Avon Products	768	345.6	119,439.36
L'Oreal	429	6.6	43.56
Unilever U.S.	323	-99.4	9,880.36
Revlon	306	-116.4	13,548.96
Garden Botanika	286	-136.4	18,604.96
	$\sum x = 2,112$		$\sum (x - \bar{x})^2 = 161,517.2$

☛ Sample variance: $s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{161,517.2}{5 - 1} = 40,379.3$

☛ Sample standard deviation: $s = \sqrt{s^2} = \sqrt{40,379.3} = 200.946$

(c) Find P_{60} and P_{20} .

Percentile	Percentile Position i	Percentile Value
♦ P_{60}	$i = 60\% \times 5 = 3$	$P_{60} = \frac{323 + 429}{2} = \mathbf{376}$
♦ P_{20}	$i = 20\% \times 5 = 1$	$P_{20} = \frac{286 + 306}{2} = \mathbf{296}$

(d) Find the range, interquartile range.

✎ Range = Max – Min = 768 – 286 = 482		
Percentile	Percentile Position i	Percentile Value
♦ $Q_1 = P_{25}$	$i = 25\% \times 5 = 1.25 \Rightarrow 2^{nd}$ position	$Q_1 = P_{25} = 306$
♦ $Q_3 = P_{75}$	$i = 75\% \times 5 = 3.75 \Rightarrow 4^{th}$ position	$Q_3 = P_{75} = 429$
✎ IQR = $Q_3 - Q_1 = 429 - 306 = \mathbf{123}$		

(e) Find the z score for Unilever U.S.

✎ $z = \frac{x - \bar{x}}{s} = \frac{323 - 422.4}{200.946} = \mathbf{-0.495}$

(f) Find the coefficient of variation.

✎ $CV = \frac{s}{\bar{x}} = \frac{200.946}{422.4} = 0.4757 = \mathbf{47.57\%}$
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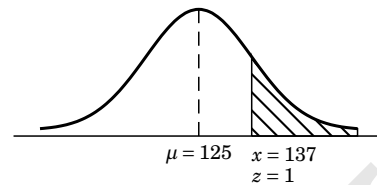
Question 3:

A distribution of numbers is approximately bell shaped. If the mean of the numbers is 125 and the standard deviation is 12, use the Empirical Rule to answer the following:

(a) *what proportion of values would be greater than 137 ?*

$$z = \frac{x - \mu}{\sigma} = \frac{137 - 125}{12} = 1$$

$$\text{Approximately } 50\% - \frac{68\%}{2} = 16\%$$



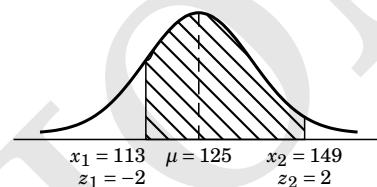
(b) *what proportion of values would fall between 113 and 149 ?*

The z-scores for $x_1 = 113$ marks and $x_2 = 149$ marks are

$$z_1 = \frac{x_1 - \mu}{\sigma} = \frac{113 - 125}{12} = -1$$

$$z_2 = \frac{x_2 - \mu}{\sigma} = \frac{149 - 125}{12} = 2$$

$$\text{Approximately } \frac{68\%}{2} + \frac{95\%}{2} = 81.5\%$$

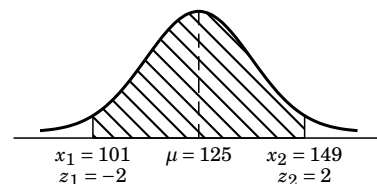


(c) *between what two numbers would approximately 95% of the values fall?*

Approximately 95% $\Rightarrow z = \pm 2$

$$z_1 = \frac{x_1 - 125}{12} = -2 \Rightarrow x_1 = 101$$

$$z_2 = \frac{x_2 - 125}{12} = 2 \Rightarrow x_2 = 149$$



Question 4:

Some numbers are not normally distributed. If the mean of the numbers is 38 and the standard deviation is 6,

- (a) *what proportion of values would fall between 26 and 50? What theorem are you using to justify your answer?*

$$\bullet k_1 = \frac{26 - 38}{6} = -2$$

$$\bullet k_2 = \frac{50 - 38}{6} = 2$$

✎ According to the Chebyshev's Theorem, **at least** $1 - \frac{1}{k^2} = 1 - \frac{1}{2^2} = \boxed{75\%}$ fall between 26 and 50.

- (b) *between what two values would at least 89% of the values fall?*

$$\text{✎ At least } 1 - \frac{1}{k^2} = 89\% \Rightarrow k = \pm 3$$

$$\bullet k_1 = \frac{x_1 - 38}{6} = 20 \Rightarrow x_1 = \boxed{20}$$

$$\bullet k_2 = \frac{x_2 - 38}{6} = 56 \Rightarrow x_2 = \boxed{56}$$

✎ At least 89% fall between 20 and 56.

Question 5:

The number of accidents that occur on an assembly line have a Poisson distribution, with an average of three accidents per week.

- (a) Find the probability that a particular week will be accident free.

$$P(x = 0) = \mathbf{0.0498}$$

- (b) Find the probability that at least three accidents will occur in a week.

$$P(x \geq 3) = 1 - P(x < 3) = 1 - 0.0498 - 0.1494 - 0.2240 = \mathbf{0.5768}$$

- (c) Find the probability that exactly five accidents will occur in a week.

$$P(x = 5) = \mathbf{0.1008}$$

- (d) Find the expected number of accidents in a year.

$$\hookrightarrow \text{New } \lambda = 3 \times 52 = 156 \text{ accidents in a year}$$

$$\hookrightarrow E(x) = \mu = \lambda = \mathbf{156} \text{ accidents}$$

- (e) Find the probability that at most 4 accidents occur in a fortnight.

$$\hookrightarrow \text{New } \lambda = 3 \times 2 = 6 \text{ accidents in a fortnight}$$

$$P(X \leq 4) = 0.0025 + 0.0149 + 0.0446 + 0.0892 + 0.1339 = \mathbf{0.2851}$$

Question 6:

A shoe store's records show that 30% of customers making a purchase use a credit card to make payment. This morning, 20 customers purchased shoes from the store. Assuming that the number of customers making a purchase using a credit card follows a Binomial distribution, answer the following questions:

- (a) What is the probability that exactly 5 of the customers used a credit card?

$$\hookrightarrow n = 20; \text{ Chance of success: } p = 0.3$$

$$\hookrightarrow P(x = 5) = \mathbf{0.179}$$

- (b) What is the probability that at most 3 of the customers used a credit card?

$$P(x \leq 3) = 0.001 + 0.007 + 0.028 + 0.072 = \mathbf{0.108}$$

- (c) What is the probability that at least three customers, but not more than six, used a credit card?

$$P(3 \leq X \leq 6) = 0.072 + 0.130 + 0.179 + 0.192 = \mathbf{0.573}$$

- (d) What is the expected number of customers who used a credit card? What's the standard deviation?

$$\hookrightarrow n = 20; \text{ Chance of success: } p = 0.3$$

$$\hookrightarrow E(\text{not } x) = \mu = np = 0.3 \times 20 = \mathbf{6} \text{ (customers)}$$

$$\hookrightarrow \sigma = \sqrt{npq} = \sqrt{20 \times 0.3 \times 0.7} = \mathbf{2.0494} \text{ (customers)}$$

- (e) What is the expected number of customers who did not use a credit card? What's the standard deviation?

$$\hookrightarrow n = 20; \text{ Chance of success: } p = 0.3$$

$$\hookrightarrow E(\text{not } x) = \mu = np = 0.7 \times 20 = \mathbf{14} \text{ (customers)}$$

$$\hookrightarrow \sigma = \sqrt{npq} = \sqrt{20 \times 0.7 \times 0.3} = \mathbf{2.0494} \text{ (customers)}$$

Question 7:

In a survey of 40 stocks (in total) in the aerospace sector and in the food sector, the stock brokers regarded 19 stocks as having a risk rated as "high". Of the 15 stocks in the aerospace sector, 6 were rated as being "low risk".

☛ Your calculation space:

This part won't be marked

	A	F	Total
H	9/40	10/40	19/40
L	6/40	15/40	21/40
Total	15/40	25/40	40/40

If a stock is chosen at random, what is the probability that:

(a) *the company is low risk?*

$$P(L) = \frac{21}{40} = \mathbf{0.525}$$

(b) *the company is in the food sector and and low risk?*

$$P(F \cap L) = \frac{15}{40} = \mathbf{0.375}$$

(c) *the company is in the aerospace sector or has high risk?*

$$P(A \cup H) = P(A) + P(H) - P(A \cap H) = \frac{15}{40} + \frac{19}{40} - \frac{9}{40} = \mathbf{0.625}$$

(d) *Given that a company is in the aerospace sector, what is the probability that it is low risk?*

$$P(L|A) = \frac{P(L \cap A)}{P(A)} = \frac{6/40}{15/40} = \mathbf{0.4}$$

Question 8:

The random variable x is uniformly distributed over a range of values from 8 to 21.

(a) What is the value of $f(x)$ for this distribution?

$$f(x) = \frac{1}{b-a} = \frac{1}{21-8} = \frac{1}{13}$$

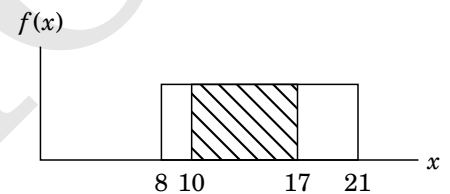
(b) Determine the average and standard deviation of this distribution.

Mean: $\mu = \frac{b+a}{2} = \frac{21+8}{2} = 14.5$

Standard deviation: $\sigma = \frac{b-a}{\sqrt{12}} = \frac{21-8}{\sqrt{12}} = 3.7528$

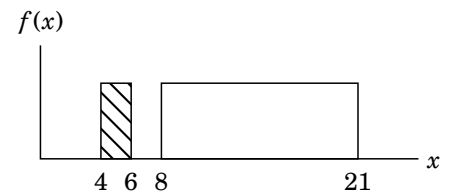
(c) $P(10 \leq x < 17) = ?$

$$P(10 \leq x < 17) = \frac{x_2 - x_1}{b - a} = \frac{17 - 10}{21 - 8} = 0.5385$$



(d) $P(4 \leq x \leq 6) = ?$

$$P(4 \leq x \leq 6) = 0$$



Question 9:

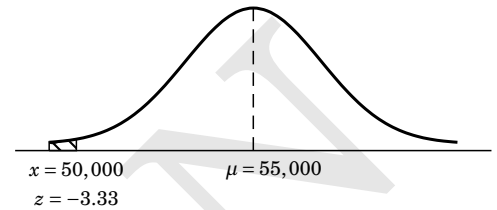
The average salary for chemists in New Zealand is NZ\$55,000. Suppose the standard deviation of these salaries is \$1,500 and that the salaries are normally distributed.

- (a) What is the probability that a new chemistry graduate will be offered a salary of less than \$50,000?

$$* \quad P(x < 50,000) = ?$$

$$\hookrightarrow z = \frac{x - \mu}{\sigma} = \frac{50,000 - 55,000}{1,500} = -3.33$$

$$\hookrightarrow P(x < 50,000) = P(z < -3.33) = 0.5 - 0.4996 = \mathbf{0.0004}$$

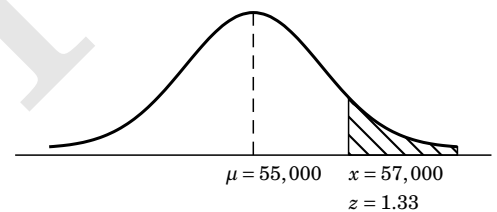


- (b) What is the probability that a new chemistry graduate will be offered a salary of more than \$57,000?

$$* \quad P(x > 57,000) = ?$$

$$\hookrightarrow z = \frac{x - \mu}{\sigma} = \frac{57,000 - 55,000}{1,500} = 1.33$$

$$\hookrightarrow P(x > 57,000) = P(z > 1.33) = 0.5 - 0.4082 = \mathbf{0.0918}$$



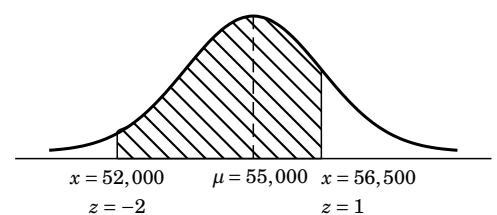
- (c) What is the probability that a new chemistry graduate will be offered a salary of between \$52,000 and \$56,500?

$$* \quad P(52,000 \leq x \leq 56,500) = ?$$

$$\hookrightarrow z = \frac{x - \mu}{\sigma} = \frac{52,000 - 55,000}{1,500} = -2$$

$$\hookrightarrow z = \frac{x - \mu}{\sigma} = \frac{56,500 - 55,000}{1,500} = 1$$

$$\hookrightarrow P(52,000 \leq x \leq 56,500) = P(-2 \leq z \leq 1) = 0.4772 + 0.3413 = \mathbf{0.8185}$$

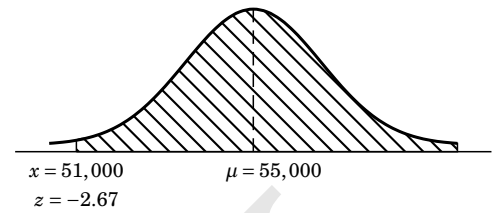


(d) What is the probability that a new chemistry graduate will be offered a salary of more than \$51,000?

$$* \boxed{P(x > 51,000) = ?}$$

$$\hookrightarrow z = \frac{x - \mu}{\sigma} = \frac{51,000 - 55,000}{1,500} = -2.67$$

$$\hookrightarrow P(x > 51,000) = P(z > -2.67) = 0.5 + 0.4962 = \mathbf{0.9962}$$



(e) Below what salary level will 90% of the new chemistry graduates be offered?

$$* \boxed{P(x < m) = 0.9}$$

$$\hookrightarrow z^* = \frac{x - \mu}{\sigma} = \frac{m - 55,000}{1,500}$$

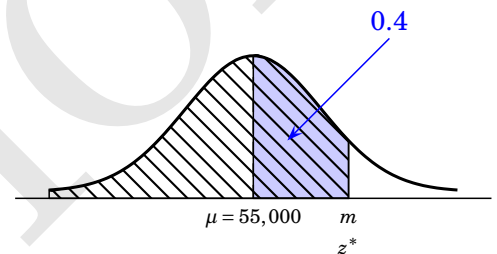
$$\hookrightarrow \text{Since } P(55,000 < x < m) = P(0 < z < z^*) = 0.9 - 0.5 = 0.4$$

$\Rightarrow$ From the standard normal table, $z^* = 1.28$

$$\hookrightarrow z^* = \frac{m - 55,000}{1,500} = 1.28$$

$$\Rightarrow m = 1.28 \times 1,500 + 55,000 = \mathbf{56,920}$$

☛ 90% of new chemistry graduates will be offered a salary of below \$56,920



Question 10:

Assume that the weights of adult Australian males are averaged at 70.3 kg with a standard deviation of 2.74 kg. (**Show your workings, including name of theorem that you use to answer the questions, if any.**)

(a) What is the mean and standard deviation of the sampling distribution of the sample means with a size of 36?

* Because the sample size $n = 36 > 30$, the sampling distribution of the sample means is approximately normally distributed according to the Central Limit Theorem.

$$\begin{cases} \mu_{\bar{x}} = \mu = 70.3 \\ \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{2.74}{\sqrt{36}} = 0.4567 \end{cases}$$

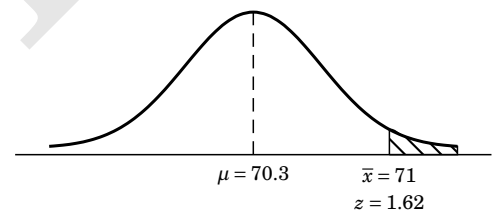
(b) If a random sample of 40 adult males is chosen, what is the probability that their mean weight is greater than 71 kg?

* $P(\bar{x} > 71) = ?$

Because the sample size $n = 40 > 30$, the sampling distribution of the sample means is approximately normally distributed according to the Central Limit Theorem.

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{71 - 70.3}{\frac{2.74}{\sqrt{40}}} = 1.62$$

$$P(\bar{x} > 71) = P(z > 1.62) = 0.5 - 0.4474 = \mathbf{0.0526}$$



(c) If a random sample of 25 adult males is chosen, what is the probability that their mean weight is less than 72 kg?

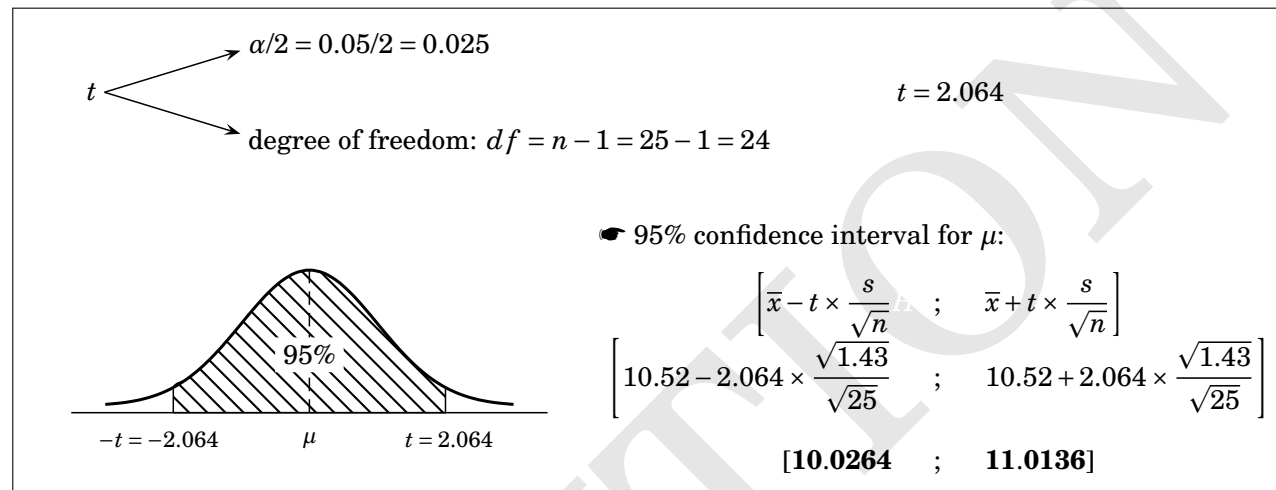
Since the population is not known to be normally distributed and the sample size $n = 25 < 30$, the sampling distribution of the sample means is not normally distributed, therefore within the scope of this course, this problem cannot be solved.

Question 11:

- (a) A NSW Department of Consumer Affairs officer responsible for enforcing laws concerning weights and measures routinely inspects packages to determine if the weight of the contents of 10 kg bags of potatoes is at least as great as that advertised on the bag. A random sample of 25 bags whose container claims that the net weight is 10 kg yielded the following statistics:

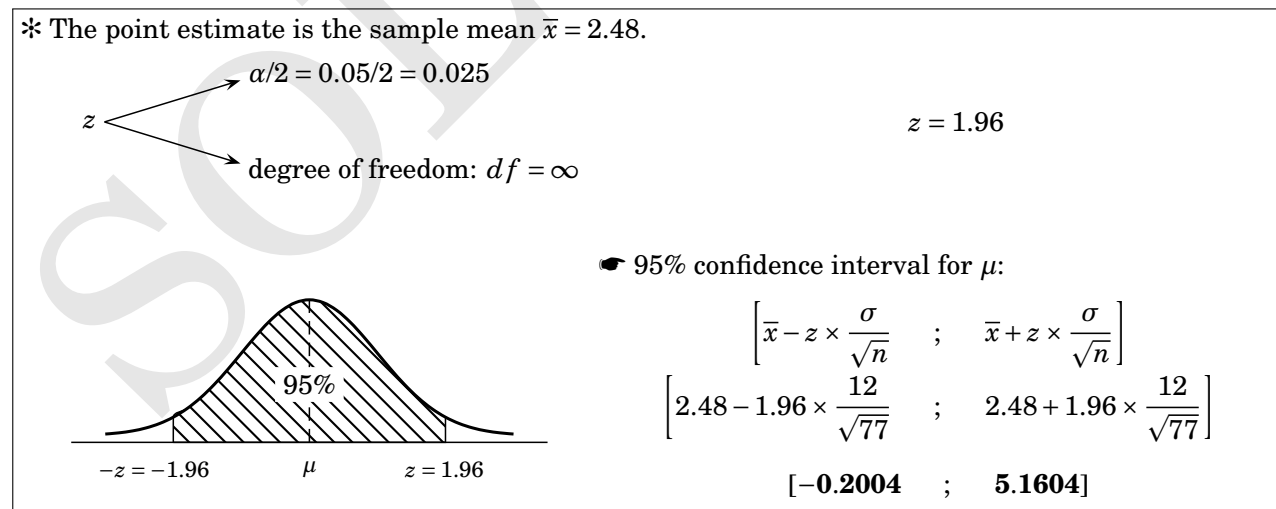
$$\bar{x} = 10.52 \text{ and } s^2 = 1.43$$

Estimate with 95% confidence the mean weight of the container assuming the underlying population is normally distributed.



- (b) A survey of 77 commercial airline flights of under 2 hours resulted in a sample average late time for a flight of 2.48 minutes. The population standard deviation was 12 minutes. What is the point estimate for the population mean? Construct a 95% confidence interval for the average time that a commercial flight of under 2 hours is late.

* The point estimate is the sample mean $\bar{x} = 2.48$.



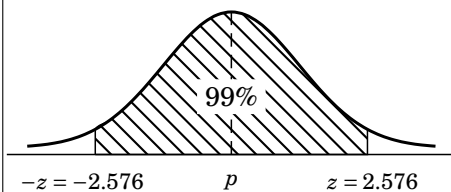
- (c) A survey of 80 students is selected randomly at Insearch. They are asked if they use an iPad to take notes in class. Based on the survey, 15 of the 80 students responded 'yes'. Identify the point estimate for the population proportion and if possible, construct a 99% interval estimate for the proportion of all Insearch students who use iPads to take notes in class.

✎ The point estimate is the sample proportion $\hat{p} = \frac{15}{80} = 0.1875$

✎ Check conditions for interval estimate: $n\hat{p} = 80 \times 0.1875 = 15 > 5$ and $n\hat{q} = 80 \times 0.8125 = 65 > 5$

z $\swarrow$ $\alpha/2 = 0.01/2 = 0.005$
 $\searrow$ degree of freedom: $df = \infty$

$$z = 2.576$$



☛ 99% confidence interval for p :

$$\left[\hat{p} - z \times \sqrt{\frac{\hat{p}\hat{q}}{n}} ; \hat{p} + z \times \sqrt{\frac{\hat{p}\hat{q}}{n}} \right]$$

$$\left[0.1875 - 2.576 \times \sqrt{\frac{0.1875 \times 0.8125}{80}} ; 0.1875 + 2.576 \times \sqrt{\frac{0.1875 \times 0.8125}{80}} \right]$$

$$[0.0751 ; 0.2999]$$

- (d) A random sample of 10 CEOs from the Forbes 500 revealed average total annual compensations of \$10 million with a sample standard deviation of \$15 million. Construct a 95% percent confidence interval estimate of the average total annual compensations of all Forbes 500 CEOs.

✎ Since the population is not known to be normally distributed as well as the sample size $n = 10 < 30$, this problem cannot be solved in the scope of this course.

Question 12: 8.51 and 8.59 from textbook (Chapter 8, p. 301 and 302)

- (a) Determine the sample size necessary to estimate, with 95% confidence, the average weight gain of rats on a special cereal diet, to within 2 g. Assume that the population standard deviation is 15 g.

✎ 95% confidence $\Rightarrow z_{0.025} = 1.960$

✎ The minimal sample size is:

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 = \left(\frac{1.960 \times 15}{2} \right)^2 = 216.09$$

✎ Since you cannot sample 216.09 units, you need to round it up to **n = 217**.

- (b) An entrepreneur wants to open an appliance service repair shop. She would like to know about the average home repair bill, including the charge for the service call for appliance repair in the area. She wants the estimate to be within \$20 of the actual figure. She believes the range of such bills is between \$30 and \$600. How large a sample should the entrepreneur take if she wants to be 95% confidence of the results?

✎ Confidence level of 95%: $z_{0.025} = 1.96$

✎ Range = $600 - 30 = 570$ points

✎ Population standard deviation $\approx \frac{\text{Range}}{4} = \frac{570}{4} = 142.5$ points

$$n = \left(\frac{z_{\alpha/2} \times \sigma}{E} \right)^2 = \left(\frac{1.96 \times 142.5}{20} \right)^2 = 195.0212$$

✎ The minimum sample size is **196 bills** (rounded up).

- (c) Suppose you want to estimate the proportion of cars that are sports utility vehicles being driven in Melbourne at rush hour by standing on the corner of Flinders Street and Swanston Street and counting sports utility vehicles. You believe the figure is no higher than 0.40. If you want the error of the confidence interval to be no greater than 0.03, how many cars should you randomly sample? Use a 90% level of confidence.

✎ Confidence level of 90%: $z_{0.05} = 1.645$

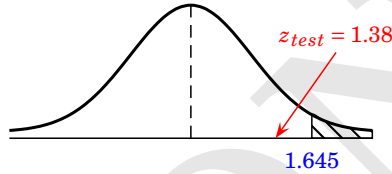
$$n = \frac{z_{\alpha/2}^2 \times pq}{E^2} = \frac{1.645^2 \times 0.4 \times 0.6}{0.03^2} = 721.6067$$

✎ You need to pick a minimum random sample of **722 cars** (rounded up).

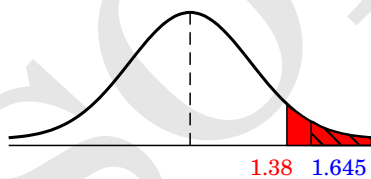
Question 13:

A life insurance salesperson claims the average worker in Perth has no more than \$25,000 of personal life insurance. To test this claim, you randomly sample 100 workers in Perth. You find that this sample of workers averages \$26,650 of personal life insurance. The population standard deviation is \$12,000.

(a) Does the salesperson's claim appear invalid? Use a level of significance of 5%.

1. State the null hypothesis State the alternative hypothesis	$H_0 : \mu \leq 25,000$ $H_a : \mu > 25,000$ 
2. Statistical test: Use z or t ?	z test since we know σ
3. Level of Significance	$\alpha = 0.05$
4. Set up the critical values z_{crit} or t_{crit}	$z_{\text{crit}} = 1.645$ Decision rule: Reject H_0 if $z_{\text{test}} > 1.645$
5. Gather sample data	$\bar{x} = 26,650; n = 100; \sigma = 12,000$
6. Calculate the test statistics z_{test} or t_{test}	$z_{\text{test}} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{26,650 - 25,000}{\frac{12,000}{\sqrt{100}}} = 1.38$
7. Make the statistical conclusion	Don't reject H_0 because $z_{\text{test}} < 1.645$ There's no evidence to suggest that the salesperson's claim is invalid.

(b) Determine p -value for the test and reach the conclusion using the p -value.



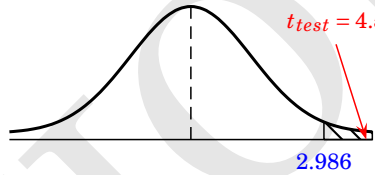
☛ $p\text{-value} = 0.5 - 0.4162 = 0.0838$

☛ Since $p\text{-value} > \alpha$, do not reject the null hypothesis H_0

Question 14:

A study of pollutants showed that certain industrial emissions should not exceed 2.5 parts per million. You believe that a particular company may be exceeding this average. To test this supposition, you randomly take a sample of nine air tests. The sample average is 3.4 parts per million, with sample standard deviation of 0.6.

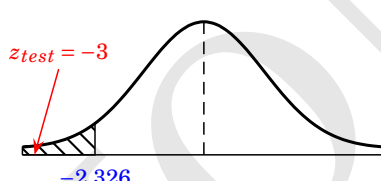
Does this result provide enough evidence for you to conclude that the company is exceeding the safe limit? The probability of committing a Type I error is assumed to be 0.01. Assume emissions are normally distributed.

1. State the null hypothesis State the alternative hypothesis	$H_0 : \mu \leq 2.5$ $H_a : \mu > 2.5$ 
2. Statistical test: Use z or t ?	t test since we don't know σ
3. Level of Significance	$\alpha/2 = 0.01 = 0.01$
4. Set up the critical values z_{crit} or t_{crit}	$t_{\text{crit}} = t_{0.01,8} = 2.896$ Decision rule: Reject H_0 if $t_{\text{test}} > 2.896$
5. Gather sample data	$\bar{x} = 3.4; n = 9; s = 0.6$
6. Calculate the test statistics z_{test} or t_{test}	$t_{\text{test}} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{3.4 - 2.5}{\frac{0.6}{\sqrt{9}}} = 4.5$
7. Make the statistical conclusion	Reject H_0 because $t_{\text{test}} > 2.986$ There's sufficient evidence to suggest that the company is exceeding the safety limit.

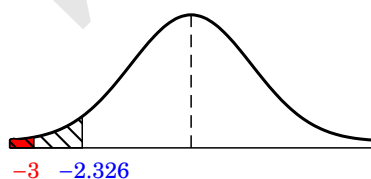
Question 15: Hypothesis Testing - 9.22 from textbook (Chapter 9, p. 337)

A study by Hewitt Associates showed that 79% of companies offer employees flexible scheduling. Suppose a researcher believes that in accounting companies this figure is lower. The researcher randomly selects 415 accounting companies and through interviews determines that 303 of these companies have flexible scheduling.

- (a) *With a 1% level of significance, does the test show enough evidence to conclude that a significantly lower proportion of accounting companies offer employees flexible scheduling?*

1. State the null hypothesis State the alternative hypothesis	$H_0 : p \geq 0.79$ $H_a : p < 0.79$ 
2. Statistical test: Use z or t ?	z test since we're testing for proportions.
3. Level of Significance	$\alpha = 0.01$
4. Set up the critical values z_{crit} or t_{crit}	$z_{\text{crit}} = -2.326$ Decision rule: Reject H_0 if $z_{\text{test}} < -2.326$
5. Gather sample data	$\hat{p} = \frac{303}{415} = 0.7301; n = 415$
6. Calculate the test statistics z_{test} or t_{test}	$z_{\text{test}} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.7301 - 0.79}{\sqrt{\frac{0.79 \times 0.21}{415}}} = -3$
7. Make the statistical conclusion	Reject H_0 because $z_{\text{test}} < -2.326$ There's sufficient evidence to suggest that a significantly lower proportion of accounting companies offer employees flexible scheduling

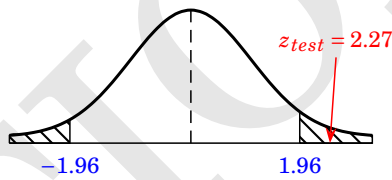
- (b) *Determine p -value for the test and reach the conclusion using the p -value.*



- $p\text{-value} = 0.5 - 0.4987 = 0.0013$
- Since $p\text{-value} < \alpha$, reject the null hypothesis H_0

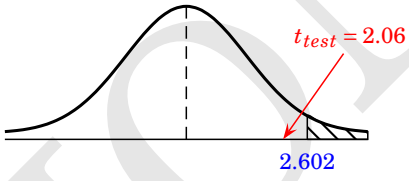
Question 16: 10.9 from textbook (Chapter 10, p. 371)

Employee suggestions can provide useful and insightful ideas for management. Some companies solicit and receive employee suggestions more than others, and company culture influences the use of employee suggestions. Suppose a study is conducted to determine whether there is a significant difference in mean number of suggestions a month per employee between company A and company B. The study shows that the average number of suggestions per month is 5.8 at company A and 5.0 at company B. Suppose these figures were obtained from random samples of 36 and 45 employees, respectively. If the population standard deviations of suggestions per employee are 1.7 and 1.4 for company A and company B, respectively, is there a significant difference in the population means? Use $\alpha = 0.05$

1. State the null hypothesis State the alternative hypothesis	$H_0 : \mu_A - \mu_B = 0$ or $\mu_A = \mu_B$ $H_a : \mu_A - \mu_B \neq 0$ or $\mu_A \neq \mu_B$ 
2. Statistical test: Use z or t ?	z test since we know σ and $n > 30$
3. Level of Significance	$\alpha/2 = 0.05/2 = 0.025$
4. Set up the critical values z_{crit} or t_{crit}	$z_{\text{crit}} = \pm 1.96$ Decision rule: Reject H_0 if $z_{\text{test}} > 1.96$ or $z_{\text{test}} < -1.96$
5. Gather sample data	Company A: $\bar{x}_A = 5.8; n_A = 36; \sigma_A = 1.7$ Company B: $\bar{x}_B = 5.0; n_B = 45; \sigma_B = 1.4$
6. Calculate the test statistics z_{test} or t_{test}	$z_{\text{test}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ $z_{\text{test}} = \frac{(5.8 - 5.0) - (0)}{\sqrt{\frac{1.7^2}{36} + \frac{1.4^2}{45}}} = 2.27$
7. Make the statistical conclusion	Reject H_0 because $z_{\text{test}} > 1.96$ There's sufficient evidence to suggest that there is a significant difference between the average number of suggestions a month per employee between company A and company B.

Question 17: 10.17 from textbook (Chapter 10, p. 380)

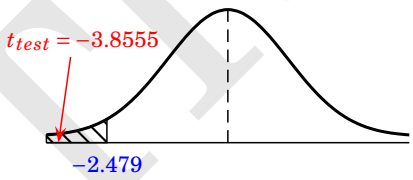
Based on an indication that mean daily car-rental rates may be higher for Melbourne than for Sydney, a survey of eight car-rental companies in Melbourne is taken and the sample mean car-rental rate is \$47, with a standard deviation of \$3. Further, suppose a survey of nine car-rental companies in Sydney results in a sample mean of \$44 and a standard deviation of \$3. Use $\alpha = 0.01$ to test whether the average daily car-rental rates in Melbourne are significantly higher than those in Sydney. Assume car-rental rates are normally distributed and the population variances are equal.

1. State the null hypothesis State the alternative hypothesis	$H_0 : \mu_M - \mu_S \leq 0$ or $\mu_M \leq \mu_S$ $H_a : \mu_M - \mu_S > 0$ or $\mu_M > \mu_S$ 
2. Statistical test: Use z or t ?	t test since we don't know σ
3. Level of Significance	$\alpha = 0.01$
4. Set up the critical values z_{crit} or t_{crit}	$t_{\text{crit}} = t_{0.01, 8+9-2} = 2.602$ Decision rule: Reject H_0 if $t_{\text{test}} > 2.602$
5. Gather sample data	Melbourne: $\bar{x}_M = 47; n_M = 8; s_M = 3$ Sydney: $\bar{x}_S = 44; n_S = 9; s_S = 3$
6. Calculate the test statistics z_{test} or t_{test}	$t_{\text{test}} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $t_{\text{test}} = \frac{(47 - 44) - 0}{\sqrt{\frac{3^2(8 - 1) + 3^2(9 - 1)}{8 + 9 - 2}} \sqrt{\frac{1}{8} + \frac{1}{9}}} = 2.06$
7. Make the statistical conclusion	Do not reject H_0 because $t_{\text{test}} < 2.602$ There's insufficient evidence to suggest that the average daily car-rental rates in Melbourne are significantly higher than those in Sydney.

Question 18: 10.28 from textbook (Chapter 10, p. 391)

Lawrence and Glover published the results of a study in the Journal of Managerial Issues in which they examined the effects of accounting company mergers on auditing delay. Auditing delay is the time between a company's fiscal year-end and the date of the auditor's report. The hypothesis is that with the efficiencies gained through mergers, the length of the audit delay would decrease.

Suppose to test their hypothesis, they examined the audit delays on 27 clients of six companies before and after the merger of a companies (a span of 5 years). Suppose further that the mean difference in audit delay for these clients from before merger to after merger was a decrease in 3.71 days and the standard deviation of the difference was 5 days. Use these data and $\alpha = 0.01$ to test whether the audit delays after the merger were significantly lower than before the merger. Assume that the differences in auditing delay are normally distributed in the population.

1. State the null hypothesis State the alternative hypothesis	$H_0 : D \geq 0 \ (D = \mu_{\text{After}} - \mu_{\text{Before}})$ $H_a : D < 0$ 
2. Statistical test: Use z or t ?	Paired t -test since we're testing for dependent populations.
3. Level of Significance	$\alpha = 0.01$
4. Set up the critical values z_{crit} or t_{crit}	$t_{\text{crit}} = t_{0.01; 27-1} = -2.479$ Decision rule: Reject H_0 if $t_{\text{test}} < -2.479$
5. Gather sample data	$\bar{d} = -3.71; s_d = 5; n = 27$
6. Calculate the test statistics z_{test} or t_{test}	$t_{\text{test}} = \frac{\bar{d} - D}{\frac{s_d}{\sqrt{n}}} = \frac{-3.71 - 0}{\frac{5}{\sqrt{27}}} = -3.8555$
7. Make the statistical conclusion	Reject H_0 because $t_{\text{test}} < -2.479$ There is evidence to conclude that the audit delays after the merger were significantly lower than before the merger.

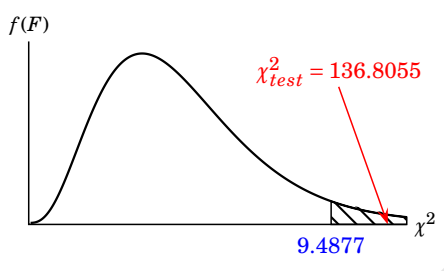
Question 19: Chi-square independence test

A recent survey has been conducted to study about how the iPads are being used by different users. 1,000 people including business people, students and general public were asked what tasks they perform the most on their iPads. The data revealed the top three tasks performed most on iPads were web surfing, social networking and media consumption. The detailed data are shown in the table below. Is there sufficient evidence to suggest the tasks performed most on iPads depend on the user types? Use a 5% level of significance.

OBSERVED FREQUENCY (f_o)				
	Web surfing	Networking	Media	TOTAL
Business people	170	80	110	360
Students	60	170	70	300
General public	80	100	160	340
TOTAL	310	350	340	1,000

EXPECTED FREQUENCY (f_e)				
	Web surfing	Networking	Media	TOTAL
Business people	111.6	126	122.4	360
Students	93	105	102	300
General public	105.4	119	115.6	340
TOTAL	310	350	340	1,000

	Expected frequency	Observed frequency	$\frac{(f_o - f_e)^2}{f_e}$
	(f_e)	(f_o)	
Business - Web surfing	111.6	170	30.5606
Business - Networking	126	80	16.7937
Business - Media	122.4	110	1.2562
Students - Web surfing	93	60	11.7097
Students - Networking	105	170	40.2381
Students - Media	102	70	10.0392
General public - Web surfing	105.4	80	6.1211
General public - Networking	119	100	3.0336
General public - Media	115.6	160	17.0533
	1,000	1,000	136.8055

1. State the null hypothesis State the alternative hypothesis	H_0 : Tasks on iPads are independent of user types H_a : Tasks on iPads are not independent of user types 
2. Statistical test	χ^2 (chi-square test of independence)
3. Level of Significance	$\alpha = 0.05$
4. Set up the critical values	$\chi^2_{\text{crit}} = \chi^2_{0.05; (3-1) \times (3-1)} = 9.4877$ Decision rule: Reject H_0 if $\chi^2_{\text{test}} > 9.4877$
5. Gather sample data	Write f_o and f_e in the given table
6. Calculate the test statistics	$\chi^2_{\text{test}} = \sum \frac{(f_o - f_e)^2}{f_e} = 136.8055$
7. Make the statistical conclusion	Reject H_0 because $\chi^2_{\text{test}} > 9.4877$ There's evidence to suggest that tasks performed on iPads are not independent of user types.

Question 20:

You want to assist a moving company owner to develop a more accurate method of predicting the labor hours needed for a moving job by using the volume of goods (in cubic feet) that is being moved. The manager has collected the following data for 36 moves and has eliminated the travel-time portion of the time needed for the move.

Spreadsheet regression results for this study are as follows:

Regression Analysis for Moving Company Study								
Regression Statistics								
Multiple R	0.9430							
R Square	0.8892							
Adjusted R Square	0.8860							
Standard Error	5.0314							
Observations	36							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	6910.7189	6910.7189	272.9864	0.0000			
Residual	34	860.7186	25.3153					
Total	35	7771.4375						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-2.3697	2.0733	-1.1430	0.2610	-6.5830	1.8437	-6.5830	1.84371
Cubic Feet Moved	0.0501	0.0030	16.5223	0.0000	0.0439	0.0562	0.0439	0.05624

(a) Identify the independent and dependent variables.

Dependent variable is "labor hours needed" and independent variable is "volume of goods (in cubic feet) that is being moved".

(b) Is the independent variable significant at $\alpha = 0.05$?

Yes, since its p -value is less than $\alpha = 0.05$.

(c) Determine and interpret the coefficient of the slope.

Slope = 0.0501 means for every increase of 1 cubic feet in the goods being moved, there is an average increase of 0.0501 labours needed.

(d) Predict the labour hours needed to finish a job to move 100 cubic feet of goods.

Predicted labour hours = $-2.3697 + 0.0501 \times 100 = 2.6403$ hours

Question 21:

Below is Excel output from a regression analysis to predict y from x_1, x_2, x_3, x_4 and x_5 .

SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.9700				
R Square	0.9409				
Adjusted R Square	0.9224				
Standard Error	3.0544				
Observations	22				
ANOVA					
	SS	df	MS	F	Significance F
Regression	2374.878	5	474.9755	50.9118	2.92E-09
Residual	149.2701	16	9.329379		
Total	2524.148	21			
	Coefficients	Standard Error	t Stat	P-value	
Intercept	-121.246	21.0916	-5.7485	2.992E-05	
x1	0.7142	0.4657	1.5336	0.1447	
x2	-0.0287	0.2239	-0.1280	0.8997	
x3	0.7308	0.1413	5.1737	9.2356E-05	
x4	0.3919	0.1203	3.2560	0.00496	
x5	0.8416	0.3283	2.5634	0.0208	

(a) Perform hypothesis testing of the overall model at $\alpha = 0.05$

$$H_o : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

H_a : At least one of the regression coefficients is $\neq 0$

Since Significance $F = 2.92E-09 < \alpha = 0.05$, reject H_o .

At least one of the coefficients is $\neq 0$.

(b) Identify predictors which are significant at $\alpha = 0.05$

Since x_3, x_4 and x_5 have p -values less than $\alpha = 0.05$, they are significant predictors.

(c) Determine and interpret the coefficient of determination of the model.

Coefficient of determination $R^2 = 92.24\%$, meaning 92.24% of the variability of dependent variable y can be explained by the independent variables.